



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Course Syllabus

Part A: Introduction

1	Course Title	Data Structure and Algorithms - I
2	Course Code	CSE 2215
3	Pre-requisites	CSE 1111, CSE 1115
4	Course Type	Core Course
5	Credit Hours	3.00
6	Section	
7	Semester	Fall 2025
8	Class Hour	
9	Course Instructor's Information	Asif Abrar (AsAr) asifabrar@cse.uiu.ac.bd Office: 933D
	Counselling Hours	Check ELMS
10	Course Rationale	This course has been designed to provide a solid foundation about the data structure and algorithms used in computer science. This course will give insights about the pros and cons of different data structures and algorithms.
11	Course Objectives	The objectives of this course are: • To familiarize the basic data structures (array, linked list). • To familiarize complex data structures (queue, stack, priority queue) using basic data structures • To use suitable data structures for different algorithms • To introduce the algorithms and their complexity and use cases

Part B: Content of the Course

12	Course Contents (approved by UGC)	Complexity Analysis: Big Oh, Big Omega, Lower and Upper Limit, Best and Worst Cases, Arrays: Memory Mapping, Vector Implementation, Sorting, Linear Time Sort, Merge Sort, Quick Sort, Linked Lists: Single Linked List, Double Linked List, Circular Linked List, Stack: Implementation using Array and Linked Lists, Application, Tower of Hanoi, Postfix Expression, Queue: Introduction to Queue, Implementation of Queue using Arrays and Linked Lists, Applications, General Tree: Introduction, Definition, Binary Tree: Introduction, Implementation using Array and Pointers, Tree Traversal Algorithms, Binary Search Tree: Operations, Insertion, Deletion, Properties, Tree Applications, Heap and Priority Queue: Introduction, Heap Sort and Application of Priority Queue, Graph Representation: Implementation using Adjacency Matrix and Adjacency List, Graphs: BFS and DFS using Adjacency Matrix and Adjacency Lists, Application of Graphs, Search Algorithms, Set Operations: Make set, Find, Union.
13	Course Outcomes (COs)	

			COs	Description											
			CO1	Able to choose appropriate data structure as applied to specified problem definition.											
			CO2	Able to handle operations like searching, insertion, deletion, traversing mechanism etc. on various data structures.											
			CO3	Able to use linear and non-linear data structures like stacks, queues, linked list etc.											
14	Mapping of COs and Program outcomes														
			COs	Program Outcomes(POs)											
				PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
			CO1			C									
			CO2			C									
			CO3				C								
15	Mapping COs with Teaching-Learning and Assessment Strategy														
			Class/ Week	Topics/Assignments			Course Outcomes (COs)	Reading Reference	Teaching- Learning Strategies	Assessment Strategies					
			1	Introduction, Course Overview				Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam					
			2	Complexity Analysis, Big Oh, Big Omega, Lower and Upper Limit, Best and Worst Cases			CO1	Lecture Slidesand Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam					
			3	Arrays: Memory Mapping, Linear & Binary Search			CO2, CO3	Lecture Slidesand Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam					
			4	Arrays: Sorting, Linear Time Sort (Count Sort)			CO1	Lecture Slidesand Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam					
			5	Arrays: Sorting, Merge Sort, Quick Sort			CO1	Lecture Slidesand Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam					
			6	Linked Lists: Single Linked List			CO2, CO3	Lecture Slidesand Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam					
			7	Linked Lists: Double Linked List, Circular Linked List			CO2, CO3	Lecture Slidesand Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam					

	8	Linked Lists: Applications, Use of slow and fast pointer	CO1	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
	9	Stack: Implementation using Array and Linked Lists	CO2, CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
	10	Stack: Application, Tower of Hanoi, Infix to Postfix, Postfix Expression Evaluation	CO1	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
	11	Queue: Implementation using Arrays and Linked lists, Applications	CO2, CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
	12	Review	CO1, CO2, CO3	Lecture Slides and Text/ Ref. Book	Lecture	Class Tests/Assignments /Quizzes/Exam
	MID EXAM					
	13	General Tree, Binary Tree: Implementation using Array and Pointers	CO2, CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
	14	Tree Traversal Algorithms, Tree Applications	CO2, CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
	15	Binary Search Tree: Operations, Insertion, Deletion, Properties, Tree Applications	CO1, CO2, CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
	16	Heap and Priority Queue: Introduction	CO1	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
	17	Heap Sort and Application of Priority Queue	CO1, CO2, CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
	18	Graph Representation: Implementation using Adjacency Matrix and Adjacency List	CO2, CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
	19	Graphs: BFS	CO2, CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
	20	Graphs: DFS	CO2, CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam

21	Graph Applications, Directed Acyclic Graph, Topological Sort	CO1, CO2, CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
22	Disjoint Set: Introduction	CO1, CO2	Lecture Slides and Text/ Ref. Book/ Online Materials	Lecture/Group Discussion/ Exercise	Class Tests/Assignments /Quizzes/Exam
23	Disjoint Set: Set Operations (Make set, Find, Union)	CO1, CO2	Lecture Slides and Text/ Ref. Book/ Online Materials	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/ Quizzes/Exam
24	Review	CO1, CO2, CO3	Lecture Slides and Text/ Ref. Book	Lecture/Group Discussion/ Exercise	Class Tests/Assignments/ Quizzes/Exam
FINAL EXAM					

Part C: Assessment and Evaluation Methods

Assessment Strategy	Assessment Types	Marks
Formative Assessment	Attendance	5%
	Assignments	5%
	Class Tests	20%
Summative Assessment	Mid Term	30%
	Final Exam	40%

Grading System

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Part D: Learning Resources

Text Book	1. Introduction to Algorithms – Thomas H. Cormen (4th edition, MIT Press & McGraw Hill, 2022) 2. Data Structures – E. M. Reingold (Addison Wesley Publication, 1998)
Reference	1. Data Structure and Algorithms in C++ - Goodrich, Tamassia (2nd edition, John Wiley and Sons Inc., 2003)

Appendix-1: Program outcomes

POs	Program Outcomes
PO1	An ability to apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

PO2	Identify, formulate, research and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.
PO3	An ability to design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and of cultural, societal and environmental concerns.
PO4	An ability to conduct investigations of complex problems, considering experimental design, data analysis and interpretation and information synthesis to provide valid conclusions.
PO5	An ability to create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex engineering activities with an understanding of their limitations
PO6	An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.
PO8	An ability to apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice.
PO9	An ability to function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.
PO10	An ability to communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PO11	An ability to demonstrate knowledge and understanding of engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.
PO12	An ability to recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.